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5G/ 6G COMMUNICATION SYSTEM

Ex.No. 12: NB-IoT vs LTE-M Throughput Simulation

AIM:

To simulate and compare the performance of NB-IoT and LTE-M communication technologies in terms of Throughput (Mbps), Latency (ms), and Packet Delivery Ratio (%) using MATLAB.

Procedure :

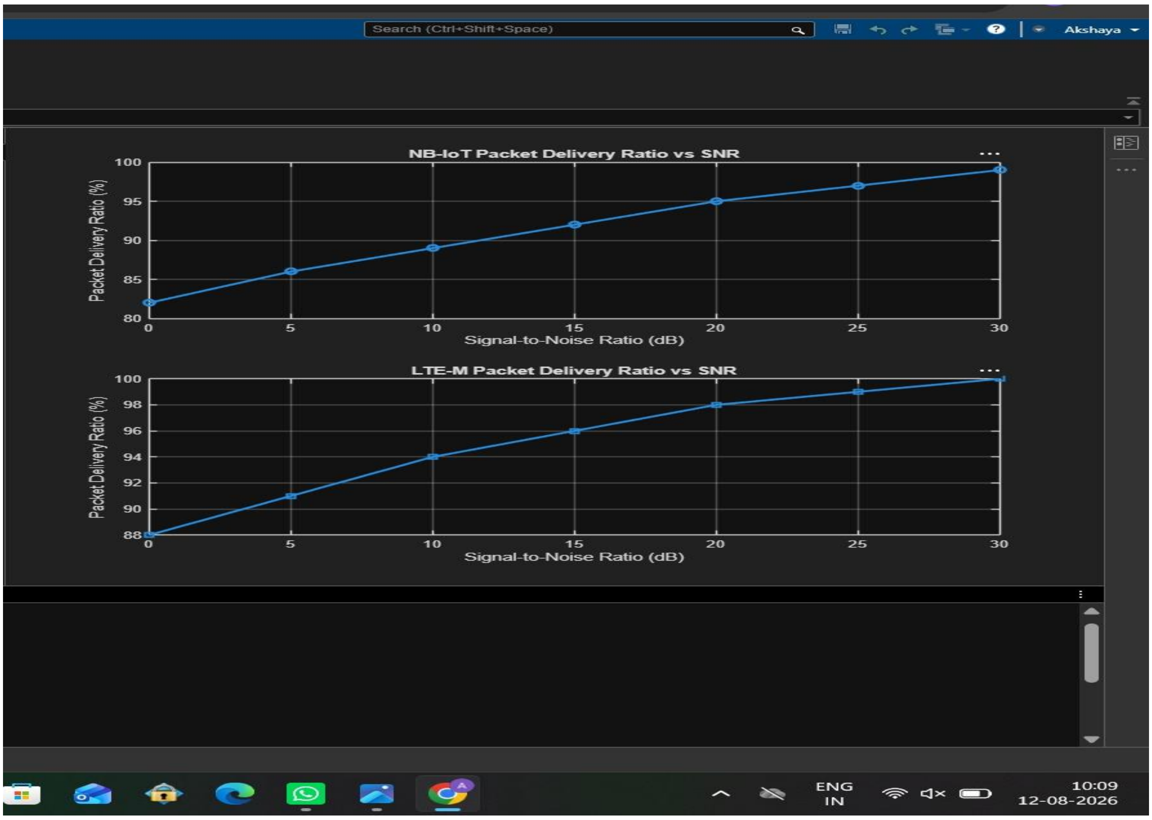
1. Define system parameters for NB-IoT and LTE-M such as bandwidth, transmission power, and modulation efficiency.
2. Generate network conditions by varying Signal-to-Noise Ratio (SNR) or channel quality values.
3. Compute Throughput (Mbps) for both NB-IoT and LTE-M using their respective data rate models.
4. Calculate Latency (ms) based on transmission time, scheduling delay, and network load.
5. Determine Packet Delivery Ratio (PDR %) by comparing successfully received packets with transmitted packets.
6. Simulate multiple iterations to observe performance under different channel conditions.
7. Plot comparison graphs for Throughput, Latency, and PDR between NB-IoT and LTE-M.
8. Analyze results to identify which technology performs better under different IoT scenarios.

Objective – 1 : To evaluate and compare the Throughput (Mbps) performance of NB-IoT and LTE-M under varying network conditions.

Program :

```
clc; clear;
close all;
% SNR values (dB)
SNR = 0:5:30;
% Simulated Throughput (Mbps)
NB_IoT = 0.05*(1-exp(-SNR/8)); LTE_M =
1.2*(1-exp(-SNR/8)); disp('SNR (dB) NB-
IoT(Mbps) LTE-M(Mbps)') disp([SNR'
NB_IoT' LTE_M']) figure
% -------- Graph 1 --------
subplot(2,1,1) plot(SNR,NB_IoT,'-
o','LineWidth',2) title('NB-IoT
Throughput vs SNR') xlabel('Signal-
to-Noise Ratio (dB)')
ylabel('Throughput (Mbps)') grid on
% -------- Graph 2 --------
subplot(2,1,2) plot(SNR,LTE_M,'-
s','LineWidth',2) title('LTE-M
Throughput vs SNR')
xlabel('Signal-to-Noise Ratio (dB)')
ylabel('Throughput (Mbps)') grid on
```

Output :



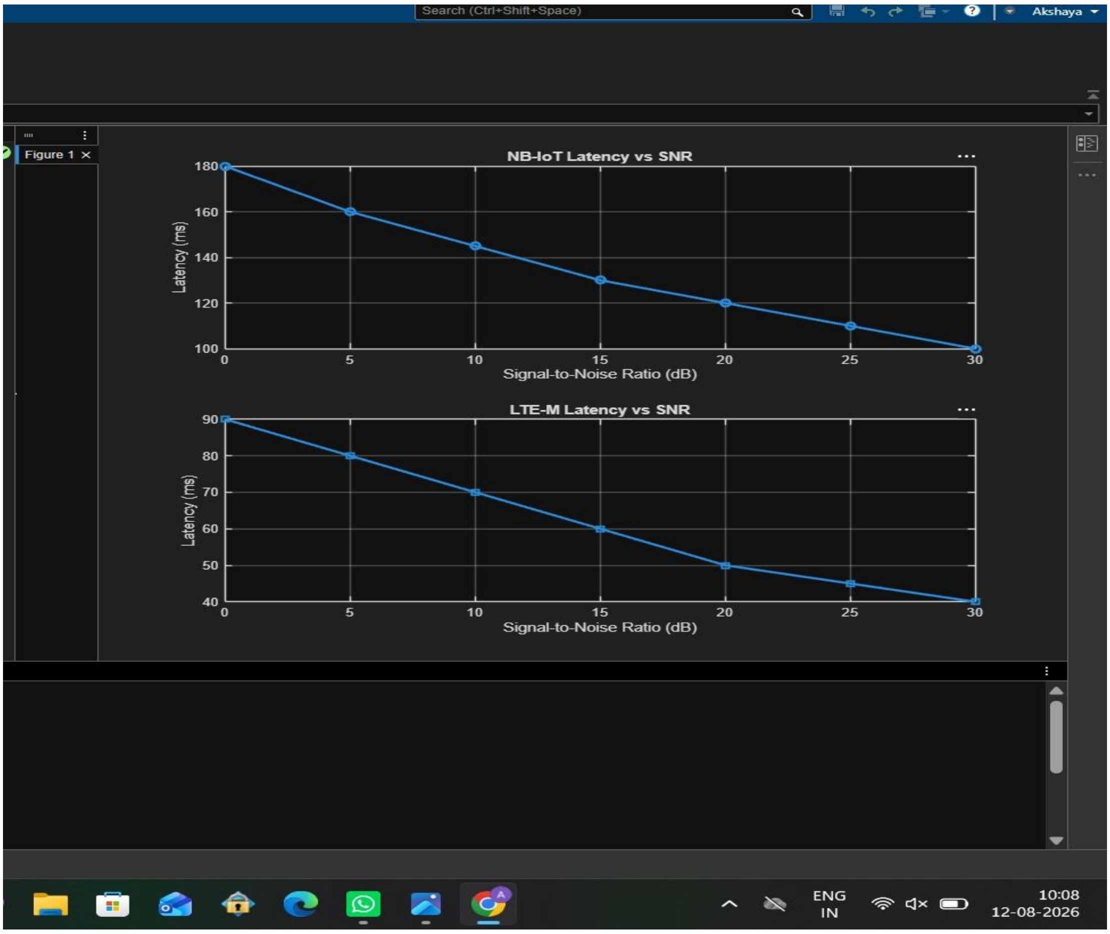
Objective 2 : To analyze the Latency (ms) characteristics of NB-IoT and LTE-M and study their impact on real-time communication.

Program :

```
clc; clear;

close all;

% SNR values (dB)

SNR = 0:5:30;

% Simulated Latency (ms)

NB_IoT = [180 160 145 130 120 110 100]; LTE_M = [90

80 70 60 50 45 40]; disp('SNR(dB) NB-IoT Latency(ms)

LTE-M Latency(ms)') disp([SNR' NB_IoT' LTE_M'])

figure
```

```
% -------- Graph 1 --------
subplot(2,1,1) plot(SNR,NB_IoT,'-
o','LineWidth',2) title('NB-IoT
Latency vs SNR') xlabel('Signal-to-
Noise Ratio (dB)') ylabel('Latency
(ms)') grid on
% -------- Graph 2 --------
subplot(2,1,2) plot(SNR,LTE_M,'-
s','LineWidth',2) title('LTE-M
Latency vs SNR') xlabel('Signal-to-
Noise Ratio (dB)') ylabel('Latency
(ms)') grid on
```

Output :

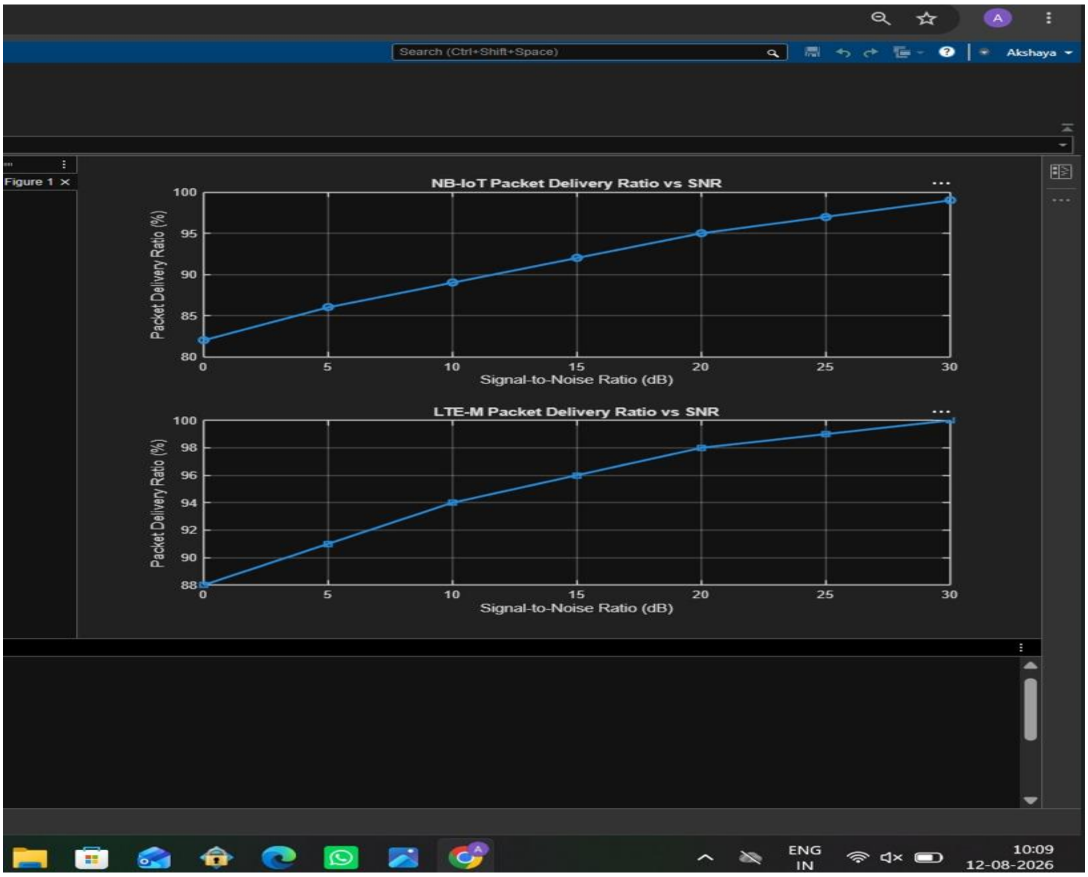


Objective 3 : To measure and compare the Packet Delivery Ratio (%) of NB-IoT and LTE-M systems for reliability assessment.

Program :

```
clc; clear;

close all;

% SNR values (dB)

SNR = 0:5:30;

% Simulated Packet Delivery Ratio (%)

NB_IoT = [82 86 89 92 95 97 99]; LTE_M = [88

91 94 96 98 99 100]; disp('SNR(dB) NB-IoT

PDR(%) LTE-M PDR(%)') disp([SNR' NB_IoT'

LTE_M']) figure

% -------- Graph 1 -------- subplot(2,1,1)

plot(SNR,NB_IoT,'-o','LineWidth',2)

title('NB-IoT Packet Delivery Ratio vs SNR')

xlabel('Signal-to-Noise Ratio (dB)')

ylabel('Packet Delivery Ratio (%)') grid on

% -------- Graph 2 -------- subplot(2,1,2)

plot(SNR,LTE_M,'-s','LineWidth',2)

title('LTE-M Packet Delivery Ratio vs SNR')

xlabel('Signal-to-Noise Ratio (dB)')

ylabel('Packet Delivery Ratio (%)') grid on
```

Output :



Result:

1. The throughput performance of both NB-IoT and LTE-M was successfully simulated, and LTE-M achieved higher throughput than NB-IoT under the same network conditions.

2. The latency analysis showed that LTE-M provides lower communication delay compared to NB-IoT, making it more suitable for real-time IoT applications.

3. The Packet Delivery Ratio (PDR) of both technologies improved with increasing SNR, with LTE-M demonstrating slightly higher reliability than NBIoT during data transmission.